

Roll Number: \_\_\_\_\_

**Thapar Institute of Engineering & Technology**  
**Department of Electronics and Communication Engineering**

M.E. (II SEM): VLSI Design

Course Code: PVL333

Course Name: System On Chip

May 28, 2025, 9:00 AM

Time: 03 Hrs.

Max. Marks: 60

Name of Faculty: Dr. Robin Singla

**Note: Attempt any 6 questions & assume missing data, if any, suitably**

| Q. No. | Questions                                                                                                                             | Marks | CO  | BL       |
|--------|---------------------------------------------------------------------------------------------------------------------------------------|-------|-----|----------|
| Q.1    | Discuss different integration platforms in SOC design.                                                                                | (10)  | CO5 | L2<br>L3 |
| Q.2    | Explain different types of system-level power estimation and modeling approaches.                                                     | (10)  | CO3 | L2<br>L3 |
| Q.3    | With the help of a block diagram, explain the transaction-level modeling.                                                             | (10)  | CO4 | L2<br>L3 |
| Q.4    | Explain SOC design methodologies.                                                                                                     | (10)  | CO5 | L2<br>L3 |
| Q.5    | Explain in detail the software design in the SOC and the different issues associated with software design in the SOC.                 | (10)  | CO3 | L2<br>L3 |
| Q.6    | Explain SOC test scheduling and test integration.                                                                                     | (10)  | CO5 | L2<br>L3 |
| Q.7    | Write a brief note on the following:<br>a) Non-pipelined<br>b) Pipelined Transfer<br>c) Burst Transfer<br>d) Pipelined Burst Transfer | (10)  | CO3 | L2<br>L3 |

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